



Operations Research

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Solution to the Training Exercises, Sheet 1

Training Exercises

Exercise T1.1. *Mining company*

Problem

A mining company owns two different pits (or mines) where certain types of ore are mined. The mines are located in different parts of the country and have different capacities. "Breaking" results in the three types of coarse-, medium- and fine-grained ore. There is a certain demand for each type of ore.

The mining company focuses on supplying at least 12 t (tons) of coarse-grained ore, 8 t of medium-grained ore, and 24 t of fine-grained ore to a metallurgical plant each week. The operating cost for the pits is 200 GE per day for pit 1 and 160 GE for pit 2. Thereby, pit 1 produces 6 t of coarse-, 2 t of medium-, and 4 t of fine-grained ore per day, while the second pit has a daily output of 2 t of coarse-, 2 t of medium-, and 12 t of fine-grained ore.

How many days should each of the pits be worked per week to meet the company's orders in an economical manner? Formulate an LP model for the solution and graphically determine an optimal solution.

Solution

Variables

x_1 number of working days in pit 1
 x_2 number of working days in pit 2

Model

Min

$$200x_1 + 160x_2 = z$$

s.t.

$$6x_1 + 2x_2 \geq 12 \quad (1)$$

$$2x_1 + 2x_2 \geq 8 \quad (2)$$

$$4x_1 + 12x_2 \geq 24 \quad (3)$$

$$x_1, x_2 \leq 7 \quad (4)$$

$$x_1, x_2 \geq 0 \quad (5)$$

Graphical solution:

Transformation of the constraints:

$$(1) \quad 6x_1 + 2x_2 \geq 12 \Rightarrow x_2 \geq 6 - 3x_1$$

$$(2) \quad 2x_1 + 2x_2 \geq 8 \Rightarrow x_2 \geq 4 - x_1$$

$$(3) \quad 4x_1 + 12x_2 \geq 24 \Rightarrow x_2 \geq 2 - 1/3x_1$$

Transformation of the objective function:

$$z = 200x_1 + 160x_2 \Rightarrow x_2 = 1/160z - 5/4x_1$$

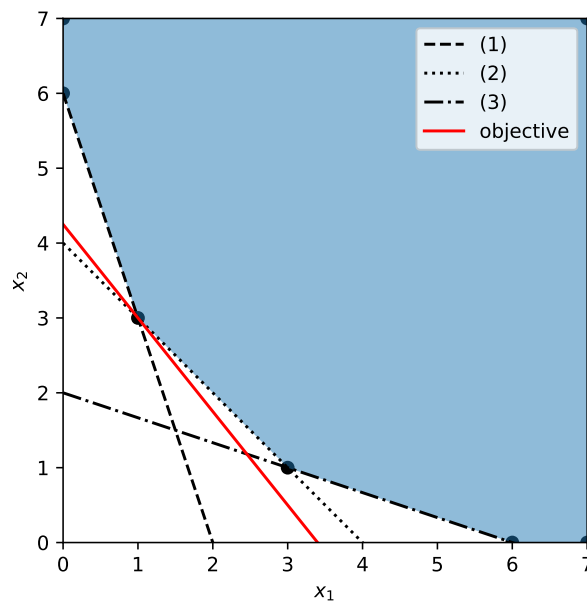
Further procedure:

- a) Enter constraints into the coordinate system
- b) Determination of the space of feasible solutions
- c) Parallel shift of the objective function (to the lower left) to the outermost point of the space of feasible solutions

The solution can be taken from the drawing:

$$x_1 = 1, x_2 = 3, z_{min} = 1 \cdot 200 + 3 \cdot 160 = 680$$

Sketch:



Exercise T1.2. Study hours

Problem

Georg studies business informatics and has to write three exams in the subjects $i = 1, \dots, 3$ at the end of the semester. There are 10 weeks left until the start of the exam period and Georg has up to 15 hours to study each week. Unfortunately, Georg realizes that his knowledge in the subjects is still limited. Without further studying he would achieve the following fictitious grades:

$\bar{F}_1$	$\bar{F}_2$	$\bar{F}_3$
8.0	3.4	6.0

Through each study session, Georg can improve his fictitious grades. For each subject i he can thereby achieve the following improvement effects v_i :

$$v_1 = \begin{cases} 0.1 & \text{for the first 30 hours} \\ 0.05 & \text{for each additional hour} \end{cases}$$

$$v_2 = \begin{cases} 0.1 & \text{for the first 6 hours} \\ 0.06 & \text{for each additional hour} \end{cases}$$

$$v_3 = \begin{cases} 0.2 & \text{for the first 5 hours} \\ 0.1 & \text{for the next 10 hours} \\ 0.05 & \text{for each additional hour} \end{cases}$$

The real final grades E_i are calculated from the fictitious grades F_i as follows:

$$E_i = \begin{cases} 5.0 & , \text{ if } F_i \geq 5.0 \\ F_i & , \text{ if } 1.0 \leq F_i \leq 5.0 \\ 1.0 & , \text{ if } F_i \leq 1.0 \end{cases}$$

George's goal is to minimize the sum of the final grades. However, it must be ensured that he passes all exams, i.e., he needs a final grade of at least 4.0 in all exams.

- a) Model this problem as a linear optimization problem.
- b) Determine an optimal solution of your LP using Gurobi. According to this, how many hours should Georg study for each subject? What would be his final grade in each subject? What is the minimum sum of his final grades?

Solution

a) *Variables*

F_i Fictitious grade achieved in subject $i = 1, \dots, 3$

E_i Final grade achieved in subject $i = 1, \dots, 3$

t_i Study time in hours for subject $i = 1, \dots, 3$

To account for the dependence of the improvement effects on the study time, we need additional auxiliary variables. For subject 1 we define the variables $t_1^{\leq 30}$ and $t_1^{>30}$ for the learning effects of the first 30 and the following hours. For subject 2 we define analogously the variables $t_2^{\leq 6}, t_2^{>6}$ and for subject 3 the variables $t_3^{\leq 5}, t_3^{>5, \leq 15}, t_3^{>15}$.

Model

$$\begin{aligned}
& \text{Min } \sum_{i=1}^3 E_i \\
& \text{s.t. } \sum_{i=1}^3 t_i \leq 150 \tag{1} \\
& t_1 = t_1^{\leq 30} + t_1^{> 30} \tag{2} \\
& t_2 = t_2^{\leq 6} + t_2^{> 6} \tag{3} \\
& t_3 = t_3^{\leq 5} + t_3^{> 5, \leq 15} + t_3^{> 15} \tag{4} \\
& F_1 = 8.0 - 0.1t_1^{\leq 30} - 0.05t_1^{> 30} \tag{5} \\
& F_2 = 3.4 - 0.1t_2^{\leq 6} - 0.06t_2^{> 6} \tag{6} \\
& F_3 = 6.0 - 0.2t_3^{\leq 5} - 0.1t_3^{> 5, \leq 15} - 0.05t_3^{> 15} \tag{7} \\
& F_i \leq 4.0 \quad \forall i \in [3] \tag{8} \\
& E_i \geq F_i \quad \forall i \in [3] \tag{9} \\
& E_i \geq 1.0 \quad \forall i \in [3] \tag{10} \\
& t_1^{\leq 30} \leq 30 \tag{11} \\
& t_2^{\leq 6} \leq 6 \tag{12} \\
& t_3^{\leq 5} \leq 5 \tag{13} \\
& t_3^{> 5, \leq 15} \leq 10 \tag{14} \\
& E_i, F_i, t_i \geq 0 \quad \forall i \in [3] \tag{15} \\
& t_1^{\leq 30}, t_1^{> 30} \geq 0 \tag{16} \\
& t_2^{\leq 6}, t_2^{> 6} \geq 0 \tag{17} \\
& t_3^{\leq 5}, t_3^{> 5, \leq 15}, t_3^{> 15} \geq 0 \tag{18}
\end{aligned}$$

Explanation:

Objective function: Minimization of the sum of the final grades

Constraints:

- (1) Time constraint: only 10 * 15 hours are available
- (2)-(4) Breakdown of study times by improvement effects
- (5)-(7) Improvement of fictitious grades
- (8) All fictitious grades must be at least 4.0
- (9) Relationship between final and fictitious grades
- (10) Final grades cannot be better than 1.0

(11)-(14) Limits for the breakdown of the time intervals

(15)-(18) Nonnegativity

Note:

Since the final grades (and hence the fictitious grades) are minimized, time units of type $t_i^>$ are used in the optimal solution only when the associated units $t_i^<$ are completely used up. A solution S in which units of type $t_i^>$ are used even though the more effective time units $t_i^<$ have not been used up to the limit can be improved by using the remaining more effective time units first; thus S would not be optimal. For this reason, no further constraints ensuring that the more effective time units are used first need to be introduced; this happens automatically.

b) Gurobi provides the following solution:

	Study time	Final grade
Subject 1	99	1.55
Subject 2	36	1.0
Subject 3	15	4.0

An alternative solution would be:

	Study time	Final grade
Subject 1	50	4.0
Subject 2	36	1.0
Subject 3	64	1.55

Due to the convexity of the linear program, there are therefore an infinite number of optimal solutions. Georg has to study at least 50 hours for subject 1 and at least 15 hours for subject 3 to pass. He can divide the remaining $150 - 50 - 36 - 15 = 49$ hours arbitrarily between subject 1 and subject 3.

The minimum sum of his final grades (= the optimal objective function value of the LP) equals 6.55 in all optimal solutions.

Exercise T1.3. *Vitamin tablets*

Problem

A manufacturer of vitamin preparations needs raw materials $R_1, \dots, R_m$ for the production of a vitamin tablet V . The cost of the raw materials is different, that is, one unit of R_k costs c_k monetary units.

The manufacturer guarantees a minimum content g_i of vitamins V_i ($i = 1, \dots, n$) per vitamin tablet. Each raw material R_k contains a proportion a_{ik} of vitamins V_i . In addition to vitamins, each raw material contains u_k units of an undesirable substance U , which is legally limited to an amount of l .

Since the manufacturer produces many different vitamin tablets, he is interested in a generic model for calculating the minimum cost of the raw materials that takes into account the

legal requirements. Design a linear program that will help you decide how many units x_k of raw materials R_k should be used to produce one vitamin tablet V . The derivation of the model must be documented in comprehensible steps.

Solution

Variables:

x_k number of units of R_k used with $k \in [m]$

Model:

Objective function:

How many units x_k of R_k should be used so that the total cost of raw materials used is minimal?

Raw materials R_k with cost c_k per unit $\Rightarrow \text{Min } z = \sum_{k=1}^m c_k x_k$

Constraints:

- a) Minimum content g_i of vitamin V_i per unit of vitamin tablet V , where each unit of raw material R_k contains a_{ik} units of V_i .

$$\Rightarrow \sum_{k=1}^m a_{ik} x_k \geq g_i \quad \forall i = 1, \dots, n$$

- b) Maximum content l of the undesirable substance U per vitamin tablet V , where the raw material R_k contains u_k units of U per unit $\Rightarrow \sum_{k=1}^m u_k x_k \leq l$

- c) Nonnegativity $\Rightarrow x_k \geq 0 \quad \forall k = 1, \dots, n$

This results in the following linear optimization model:

$$\begin{array}{llll} \text{Min} & \sum_{k=1}^m c_k x_k & = & z \\ \text{s.t.} & \sum_{k=1}^m a_{ik} x_k & \geq & g_i \quad \forall i = 1, \dots, n \\ & \sum_{k=1}^m u_k x_k & \leq & l \\ & x_k & \geq & 0 \quad \forall k = 1, \dots, m \end{array}$$

Self-study Exercises

Exercise S1.1. LP mapping

Problem

Let there be four linear programs in algebraic representation (P_1, P_2, P_3, P_4) and another four linear programs in graphical representation (G_1, G_2, G_3, G_4) . In each graphical representation, the objective function is colored blue and passes through the optimal point x^* of the linear program, which is colored red. The admissible region is colored gray.

Assign to each linear program in algebraic representation exactly one matching linear program in graphical representation, so that both describe the same underlying linear program. You may assume that for each given linear program in algebraic representation exactly one matching linear program in graphical representation is given in the problem.

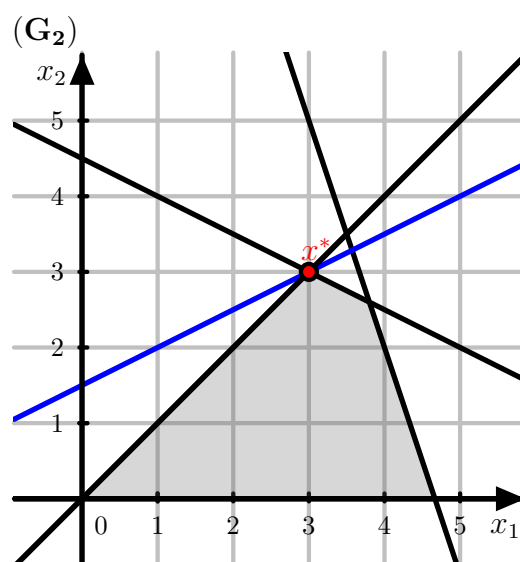
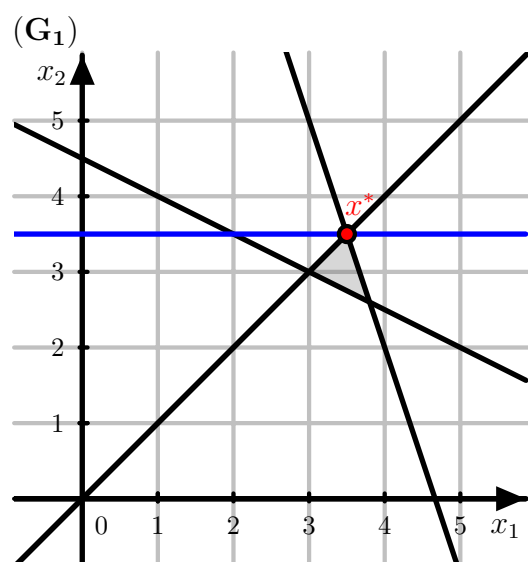
A justification for your assignment is not required.

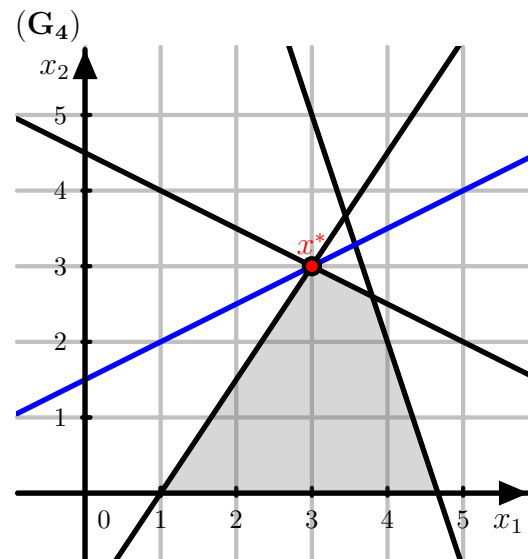
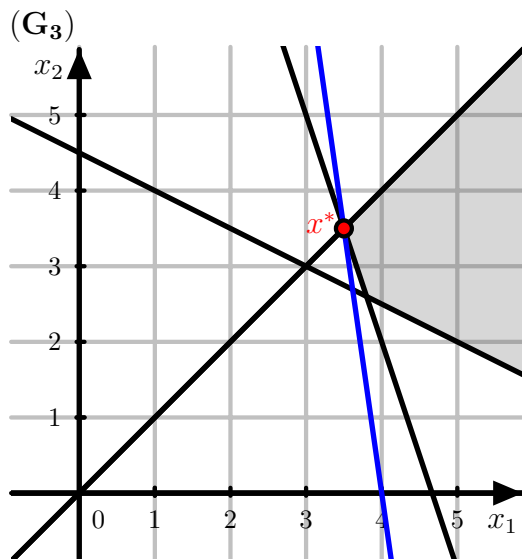
$$\begin{aligned}
 (\mathbf{P}_1) \quad & \underset{\text{s.t.}}{\text{Max}} && -x_1 + 2x_2 \\
 & x_1 + 2x_2 &\leq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_3) \quad & \underset{\text{s.t.}}{\text{Max}} && x_2 \\
 & x_1 + 2x_2 &\geq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_2) \quad & \underset{\text{s.t.}}{\text{Max}} && -x_1 + 2x_2 \\
 & x_1 + 2x_2 &\leq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -3x_1 + 2x_2 &\leq & -3 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_4) \quad & \underset{\text{s.t.}}{\text{Min}} && 7x_1 + x_2 \\
 & x_1 + 2x_2 &\geq & 9 \\
 & 3x_1 + x_2 &\geq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$





Solution

$$(P_1) \leftrightarrow (G_2)$$

$$(P_2) \leftrightarrow (G_4)$$

$$(P_3) \leftrightarrow (G_1)$$

$$(P_4) \leftrightarrow (G_3)$$

Exercise S1.2. Forecast planning

Problem

Your forecasting department provides you with the following sales forecasts, which describe how many units of your product can be sold in the first 6 months of the coming year:

m	January	February	March	April	May	June
F_m	14,000	12,000	16,000	14,000	15,000	16,000

In December, 12,000 units were produced. If the production is increased from one month to the next, there is a cost of €2 per additional unit produced. If the production is reduced from one month to the next, there is a cost of €0.50 per fewer unit produced. Overproduced units can be stored at a cost of €1 per unit in order to be sold later. At the end of December, no units are stored and at the end of June, the warehouse should also be empty.

Using a linear program, create a production plan that is cost-minimal (in terms of production and storage costs) and that covers the forecast demand.

Solution

Index range

Months $m = 0, \dots, 6$ for months December, January, ..., June

Variables

P_m Production in month m
 P_m^+ Increase in production in month m
 P_m^- Decrease in production in month m
 L_m Inventory at the end of month m

Model

$$\min \sum_{m=1}^6 (2P_m^+ + 0.5P_m^- + L_m)$$

$$\text{s.t. } L_{m-1} + P_m \geq F_m \quad \forall m = 1, \dots, 6 \quad (1)$$

$$L_m = L_{m-1} + P_m - F_m \quad \forall m = 1, \dots, 6 \quad (2)$$

$$P_m = P_{m-1} + P_m^+ - P_m^- \quad \forall m = 1, \dots, 6 \quad (3)$$

$$L_0 = 0 \quad (4)$$

$$L_6 = 0 \quad (5)$$

$$P_0 = 12,000 \quad (6)$$

$$P_m, P_m^+, P_m^-, L_m \geq 0 \quad \forall m = 1, \dots, 6 \quad (7)$$

Explanation:

Objective function: Minimization of costs

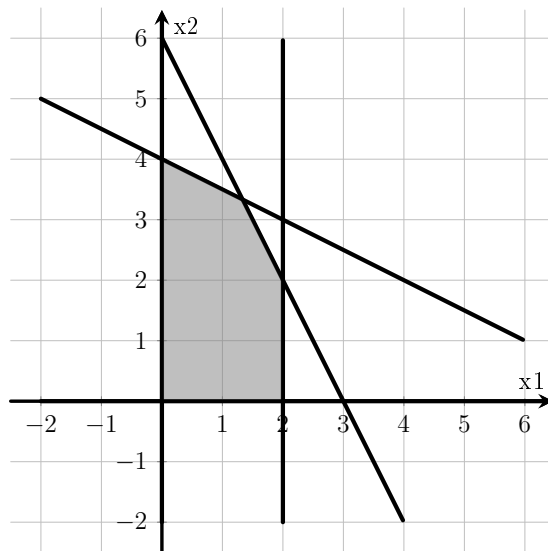
Constraints:

- (1) The number of units specified by the forecast must be achieved
- (2) Definition of the inventory level
- (3) The production in month m can be either increased or decreased compared to the previous month
- (4)-(5) Inventory condition for December and June
- (6) Production in December
- (7) Non-negativity

Exercise S1.3. *Graphical solution of an LP*

Problem

Given is the following feasible region (shaded in gray) of a linear problem.



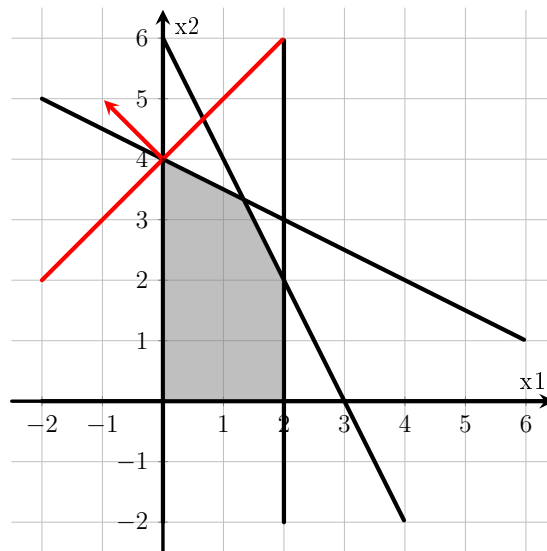
- Specify linear constraints that define this feasible region.
- Specify a vector c such that $(0, 4)$ is the *only* optimal solution of the problem $\max c^T x$ subject to the constraints from part a). Draw your objective function.
- Consider the objective vector $c' = (-1, \beta)^T$. For which choices of $\beta \in \mathbb{R}$ is the set of optimal solutions more than a single point?

Solution

- Possible inequalities:

- $0.5x_1 + x_2 \leq 4$
- $2x_1 + x_2 \leq 6$
- $x_1 \leq 2$
- $x_1, x_2 \geq 0$

- Possible solution: $c = (-1, 1)^T$



c) The only possible solution is $\beta = 0$.